

like two-wheeler or four-wheeler, transport or passenger. This representation is done using classes and each instance of a class is called an object. For e.g. Maruti 800 can be represented as an object/instance of the class 'car'. A lot of current programs and applications are based on OOP.

- In-order to strengthen your skills in OOP, work on the concepts of
- Abstraction
- Inheritance
- Encapsulation

These are the three pillars that support the entire model of OOP. You can learn these concepts in depth in your respective courses. C++ and Java are the most common OOP languages; opposed to this, C, Pascal etc. are procedural languages and they don't follow the Object Oriented approach.

By the time you are done with these concepts, you will be able to solve some basic and elementary problems. Now the main focus should be to solve complex problems and to make your code fast and efficient. Whenever a code runs on a machine, it will take up some memory and time to finish the execution. Our aim is to reduce these parameters, so that it runs quickly without taking up enormous amount of memory and time, thereby reducing the overhead on the CPU. This process is called optimization. As the problems become complex, they require more amount of time and memory, hence optimization plays a major role in deciding the quality of our code.

A lot of programming approaches have been proven and used by algorithmists so as to optimise the code. Some of these algorithms can be enlisted as Greedy Algorithm, Dynamic Programming, Divide and Conquer, Backtracking etc. These algorithms are specifically designed to solve problems, so that we require less time and memory. Often we have to compromise time over memory or vice versa. For eg: Divide and Conquer breaks the problem into smaller parts and solves them individually; later these parts are joined together to form a complete solution. It is generally preferred because of its ability to solve complex problems by breaking them down into simpler and easier problems. Similarly, each of these algorithms has their own set of advantages and disadvantages, and have a specific domain of application. If you can master these algorithms and learn their implementation in complex problems, it will be very easy to get a good job because tech-giants look for candidates who are experts in algorithms.

One of the most challenging part of a program is a bug, which can be explained as an error, flaw or an anomaly in your code, which causes unex-